

Random cubic graphs admit nonsynchronized stable Kuramoto equilibria

Abstract

For a uniformly random labelled simple cubic graph on an even number of vertices, we give a proof claim that the probability that every local minimum of the homogeneous attractive Kuramoto energy is synchronized tends to zero. This is the opposite of the limit proposed in problem MD-06 of *OpenProblemsInNLA* and Conjecture 3 of *Randomstrasse101: Open Problems of 2024*. A two-by-two sign pattern on a cycle of length divisible by four, together with a decaying nonlinear profile on its outward binary trees, has uniformly positive edge cosines and an exponentially small boundary gradient. A Laplacian spectral gap converts this truncated profile into an exact local minimum. Fixed-length cycle counts and the spectral gap of random regular graphs then establish the claimed probability limit. The analytic argument does not depend on numerical optimization. Supplementary rational existence certificates and a symbolic finite construction are checked with exact arithmetic.

Review status. Complete negative-resolution proof claim; independent mathematical review is pending. No external peer review or formal proof-assistant verification is asserted.

1 Statement and scope

Let $G = (V, E)$ be a finite simple undirected graph, and let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$. Define

$$E_G(\theta) = \sum_{\{u,v\} \in E} (1 - \cos(\theta_u - \theta_v)), \quad \theta \in \mathbb{T}^V. \quad (1)$$

A phase vector is *synchronized* when all coordinates agree modulo 2π . Local minima use the usual topology on the phase torus. Because a common phase rotation is a symmetry, a local minimum need not be strict on the entire torus.

Write L_G for the unweighted graph Laplacian, and $\lambda_2(L_G)$ for its second-smallest eigenvalue. For connected G , this eigenvalue is positive. The subspace $\mathbf{1}^\perp$ is the orthogonal complement of the constant vector in $\mathbb{R}^V$.

Theorem 1. *Let G_n be uniform over the labelled simple 3-regular graphs on $\{1, \dots, n\}$, for even $n \geq 4$. Then*

$$\lim_{\substack{n \rightarrow \infty \\ n \text{ even}}} \mathbb{P}(\text{every local minimum of } E_{G_n} \text{ is synchronized}) = 0. \quad (2)$$

More strongly, with probability tending to one, there is a nonsynchronized critical point with all edge cosines bounded below by the same positive absolute constant. Its Hessian is positive definite on $\mathbf{1}^\perp$.

The graph model, energy, phase domain, and quantification over *every* local minimum match the canonical MD-06 entry [1]. Its source is Definition 2.1 and Conjecture 3 of Bandeira, Kireeva, Maillard, and Rödder [2]. Therefore, if independently verified, Theorem 1 gives a full negative resolution, not just a family of exceptional finite graphs.

Stable nonsynchronized patterns on finite cubic graphs are not new. DeVille and Ermentrout [3] studied them analytically for specific graphs and numerically across finite graph samples. Their Section IV explicitly identifies a vanishing probability of having no stable pattern as a trend suggested by their data. The argument below is directed at the asymptotic assertion for the uniform labelled simple model; no priority claim about finite examples or the numerical phenomenon is made.

2 An exact minimum from a small gradient

We work with real lifts of the angles. The energy and its derivatives are periodic in each coordinate. Direct differentiation gives

$$(\nabla E_G(\phi))_v = \sum_{w \sim v} \sin(\phi_v - \phi_w), \quad (3)$$

$$z^\top \nabla^2 E_G(\phi) z = \sum_{\{u,v\} \in E} \cos(\phi_u - \phi_v) (z_u - z_v)^2. \quad (4)$$

The gradient has zero coordinate sum, and the Hessian annihilates $\mathbf{1}$.

Lemma 2 (Small-gradient certificate). *Suppose G is connected, $\lambda_2(L_G) \geq \gamma > 0$, and $\phi \in \mathbb{R}^V$ satisfies*

$$\cos(\phi_u - \phi_v) \geq c > 0 \quad (\{u, v\} \in E), \quad \|\nabla E_G(\phi)\|_2 < \frac{c^2 \gamma}{4\sqrt{2}}. \quad (5)$$

There is $h_ \in \mathbf{1}^\perp$, with $\|h_*\|_2 < c/(2\sqrt{2})$, such that $\theta_* = \phi + h_*$ is an exact critical point. Furthermore,*

$$\cos((\theta_*)_u - (\theta_*)_v) \geq c/2 \quad (\{u, v\} \in E), \quad (6)$$

$$z^\top \nabla^2 E_G(\theta_*) z \geq (c\gamma/2) \|z\|_2^2 \quad (z \in \mathbf{1}^\perp). \quad (7)$$

Thus θ_ is a strict local minimum modulo common rotation and a local minimum on the phase torus.*

Proof. Set $\rho = c/(2\sqrt{2})$ and $B = \{h \in \mathbf{1}^\perp : \|h\|_2 \leq \rho\}$. For each edge and each $h \in B$, $|h_u - h_v| \leq \sqrt{2} \|h\|_2 \leq c/2$. Since cosine is 1-Lipschitz, all weights in (4) at $\phi + h$ are at least $c/2$. Throughout B this implies

$$z^\top \nabla^2 E_G(\phi + h) z \geq (c/2) z^\top L_G z \geq \mu \|z\|_2^2 \quad (z \in \mathbf{1}^\perp), \quad \mu = c\gamma/2. \quad (8)$$

The continuous function $h \mapsto E_G(\phi + h)$ attains a minimum on the compact set B . At a boundary point, $\|h\|_2 = \rho$, and

$$\begin{aligned} \langle h, \nabla E_G(\phi + h) \rangle &= \langle h, \nabla E_G(\phi) \rangle + \int_0^1 h^\top \nabla^2 E_G(\phi + th) h \, dt \\ &\geq -\rho \|\nabla E_G(\phi)\|_2 + \mu \rho^2 > 0. \end{aligned}$$

A small move in the direction $-h$ decreases the energy, so a minimizer h_* is in the relative interior of B . At this point the gradient is orthogonal to $\mathbf{1}^\perp$. It is also orthogonal to $\mathbf{1}$, hence zero. The weight bound and (8) prove (6)–(7) and strict local minimality on the slice. Any nearby perturbation splits into a mean-zero perturbation and a common rotation, proving local minimality on the torus. $\square$

3 A cycle with outward binary trees

3.1 A scalar phase profile

For each integer $R \geq 4$, define

$$F_R(t) = 3 \arcsin t + 2 \sum_{j=1}^{R-1} \arcsin(t/2^j), \quad 0 \leq t \leq 1. \quad (9)$$

Lemma 3. *There is a unique $t_R \in (0, 1)$ satisfying $F_R(t_R) = 2\pi$. The roots decrease with R , and $t_R > 1/2$. Hence*

$$c_0 := \sqrt{1 - t_4^2} > 0, \quad \sqrt{1 - t_R^2} \geq c_0 \quad (R \geq 4). \quad (10)$$

Set

$$\delta_j = \arcsin(t_R/2^j), \quad a_j = \sum_{s=j}^{R-1} \delta_s \quad (0 \leq j < R), \quad a_R = 0. \quad (11)$$

Then

$$2a_0 + \delta_0 = 2\pi, \quad \sin(2a_0) = -t_R, \quad \cos(2a_0) = \sqrt{1 - t_R^2}. \quad (12)$$

Proof. Each F_R is continuous and strictly increasing, with $F_R(0) = 0$. Moreover,

$$F_4(1) = \frac{11\pi}{6} + 2 \arcsin(1/4) + 2 \arcsin(1/8) > \frac{11\pi}{6} + \frac{3}{4} > 2\pi,$$

where the last inequality follows from $\pi < 4$. Since $F_R \geq F_4$, existence and uniqueness follow by the intermediate value theorem. Increasing R increases $F_R(t)$ for $t > 0$, so the roots decrease.

Convexity of arcsine on $[0, 1]$ gives $\arcsin x \leq \pi x/2$. Using $3 \arcsin(1/2) = \pi/2$, we obtain

$$F_R(1/2) \leq \frac{\pi}{2} + \pi \sum_{j=1}^{\infty} 2^{-j-1} = \pi < 2\pi.$$

This proves $t_R > 1/2$. Finally, the identity $F_R(t_R) = 2a_0 + \delta_0$ gives (12). $\square$

Remark 4 (Explicit constants). The constant in (10) satisfies $c_0 > 1/16$. To see this, put $t_* = \sqrt{255}/16$. The elementary inequalities $\arcsin x \geq x + x^3/6$ and $\arcsin(1/16) < 1/\sqrt{255} < 1/15$ give

$$F_4(t_*) > \frac{3\pi}{2} - \frac{1}{5} + \frac{7t_*}{4} + \frac{t_*^3}{24} > \frac{3\pi}{2} + \frac{151}{96} > 2\pi.$$

Here $t_* > 99/100$, $t_*^3 > 97/100$, and $151/96 > 11/7 > \pi/2$ were used. Thus $t_4 < t_*$, proving the claim. Consequently, the final edge-cosine bound can be taken as $1/32$ and, when $\gamma = 1/10$, the Hessian bound on $\mathbf{1}^\perp$ as $(1/320)I$. In that case the explicit choice $R = 29 + \lceil \log_2 \ell \rceil$ satisfies (14), since $32\ell/((1/16)^4(1/10)^2) = 209\,715\,200\ell < 2^{28}\ell$. These conservative constants are not needed for the probability limit.

Definition 5. A cycle C in a cubic graph has a *clean radius- R neighborhood* when the subgraph induced by the vertices at distance at most R from C is unicyclic, with C as its unique cycle.

For a clean neighborhood of an ℓ -cycle, every cycle vertex has one outward child. Each vertex at depth $1, \dots, R-1$ has two children. The attached rooted trees are disjoint, and their depth- j level has $\ell 2^{j-1}$ vertices, for $1 \leq j \leq R$.

Proposition 6 (Localized approximate equilibrium). *Let $C = (v_0, \dots, v_{\ell-1})$ have length divisible by four and a clean radius- R neighborhood in a cubic graph, where $R \geq 4$. There is $\phi \in \mathbb{R}^V$ such that*

$$\min_{\{u,v\} \in E} \cos(\phi_u - \phi_v) \geq c_0, \quad \|\nabla E_G(\phi)\|_2^2 = \ell t_R^2 2^{1-R}. \quad (13)$$

On an opposite-sign cycle edge in the construction, the distance of its phase difference from $2\pi\mathbb{Z}$ is $\delta_0 > \pi/6$.

Proof. Give the cycle roots signs

$$(\varepsilon_0, \dots, \varepsilon_{\ell-1}) = (+1, +1, -1, -1, +1, +1, -1, -1, \dots).$$

Every root has one same-sign and one opposite-sign cycle neighbor. Assign phase $\varepsilon_i a_j$ to each depth- j vertex in the tree rooted at v_i , for $j < R$, including $j = 0$ for the root. Give every other vertex phase zero, including all depth- R vertices.

At a root of sign ε , the three terms in the gradient are

$$0, \quad \sin(2\varepsilon a_0) = -\varepsilon t_R, \quad \sin(\varepsilon(a_0 - a_1)) = \varepsilon t_R.$$

They cancel. At a depth- j vertex, $1 \leq j < R$, the parent contributes $-\varepsilon t_R/2^{j-1}$ and each of its two children contributes $\varepsilon t_R/2^j$. These also cancel. A depth- R vertex has phase zero, one nonzero-phase parent, and two zero-phase neighbors. Its gradient is $-\varepsilon t_R/2^{R-1}$. Beyond depth R every gradient coordinate is zero. The number of boundary vertices is $\ell 2^{R-1}$, which proves the norm identity.

Opposite-sign cycle edges have cosine $\cos(2a_0) = \sqrt{1 - t_R^2} \geq c_0$; same-sign cycle edges have cosine one. A tree edge from depth j to $j + 1$ has cosine $\cos \delta_j \geq \cos \delta_0 \geq c_0$. All other edges have phase difference zero. Lastly, a cross-sign cycle edge has difference $\pm(2\pi - \delta_0)$, where $\delta_0 = \arcsin t_R > \pi/6$. $\square$

Theorem 7 (Deterministic obstruction). *Fix $\gamma > 0$ and $\ell \in 4\mathbb{N}$, and choose an integer $R \geq 4$ with*

$$2^{R-1} > \frac{32\ell}{c_0^4 \gamma^2}. \quad (14)$$

Any connected cubic graph G with $\lambda_2(L_G) \geq \gamma$ and a clean radius- R neighborhood of an ℓ -cycle has a nonsynchronized local minimum. At this minimum every edge cosine is at least $c_0/2$ and the Hessian on $\mathbf{1}^\perp$ is bounded below by $(c_0\gamma/2)I$.

Proof. Proposition 6 and $t_R < 1$ give

$$\|\nabla E_G(\phi)\|_2 = t_R \sqrt{\ell 2^{1-R}} < \frac{c_0^2 \gamma}{4\sqrt{2}}.$$

Apply Lemma 2 with $c = c_0$. Every edge difference changes by less than $c_0/2 \leq 1/2$. On a cross-sign cycle edge, the original distance from $2\pi\mathbb{Z}$ is $\delta_0 > \pi/6 > 1/2$. Distance to a fixed closed set is 1-Lipschitz, so the new difference is not a multiple of 2π . The exact local minimum is therefore nonsynchronized. Its other properties are the conclusions of the certificate lemma. $\square$

4 Uniform random simple cubic graphs

We use two established results in their uniform labelled simple-graph forms. Bordenave's version of Friedman's theorem [4, Theorem 1, printed p. 2] bounds the nontrivial adjacency eigenvalues of a random d -regular graph by $2\sqrt{d-1} + o(1)$ in absolute value, with probability tending to one. For $d = 3$, since $3 - 2\sqrt{2} > 1/10$, it follows that

$$\mathbb{P}(\lambda_2(L_{G_n}) \geq 1/10) \longrightarrow 1. \quad (15)$$

The event in (15) also implies connectivity.

Let $C_\ell(G_n)$ count simple cycles of length ℓ . Johnson's Poisson approximation [5, Theorem 11, printed p. 12] gives, for each fixed $\ell \geq 3$,

$$C_\ell(G_n) \xrightarrow{d} \text{Poisson}(\mu_\ell), \quad \mu_\ell = \frac{2^\ell}{2\ell}. \quad (16)$$

The simple-graph convention is stated in Section 2, printed p. 3, of that paper. In particular, $\mathbb{P}(C_\ell(G_n) = 0) \rightarrow e^{-\mu_\ell}$.

Lemma 8 (Clean fixed neighborhoods). *For fixed $\ell \geq 3$ and $R \geq 1$, the probability that G_n contains an ℓ -cycle without a clean radius- R neighborhood is $O_{\ell,R}(n^{-1})$.*

Proof. A radius- R neighborhood of an ℓ -cycle in a cubic graph has at most $M = \ell 2^R$ vertices. If it is not unicyclic, it is connected and has at least two independent cycles. Repeatedly remove degree-one vertices. The remaining graph H is nonempty, has minimum degree at least two, maximum degree at most three, at most M vertices, and $e(H) - v(H) \geq 1$. Removing a leaf preserves $e - v$.

There are finitely many such isomorphism types, with a number depending only on ℓ, R . For any fixed labelled embedding of one of these types, Johnson's Proposition 1(a), printed p. 3, gives

$$\mathbb{P}(H \subseteq G_n) \leq c_1 2^{e(H)} n^{-e(H)}$$

for sufficiently large n . Its degree and size conditions hold because $d = 3$ and H is fixed. There are at most $n^{v(H)}$ embeddings of a type, so the union bound over all types is $O_{\ell,R}(n^{-1})$. $\square$

Proof of Theorem 1. Write S_n for the event that every local minimum is synchronized. Fix any $\ell \in 4\mathbb{N}$, set $\gamma = 1/10$, and choose a fixed R satisfying (14). The deterministic obstruction implies

$$\mathbb{P}(S_n) \leq \mathbb{P}(\lambda_2(L_{G_n}) < 1/10) + O_{\ell,R}(n^{-1}) + \mathbb{P}(C_\ell(G_n) = 0).$$

Indeed, if the gap condition holds, no ℓ -cycle has a dirty neighborhood, and an ℓ -cycle exists, then a nonsynchronized local minimum exists. Equations (15)–(16) yield

$$\limsup_{\substack{n \rightarrow \infty \\ n \text{ even}}} \mathbb{P}(S_n) \leq \exp\left(-\frac{2^\ell}{2\ell}\right). \quad (17)$$

This is true for *every fixed* multiple of four. Let ℓ tend to infinity after taking the limit superior; the right side tends to zero. This proves (2).

The same bound applies to the event of having no nonsynchronized minimum with edge cosines at least $c_0/2$ and Hessian lower bound $(c_0/20)I$ on $\mathbf{1}^\perp$. The constants are independent of ℓ , proving the stronger assertion. $\square$

Remark 9 (Quantifiers and scope). The events in the union bound need not be independent. In each use of a random-graph limit, both ℓ and R are fixed before n tends to infinity. No estimates uniform in growing cycle length or neighborhood radius are required. The radius bound is conservative, so this proof does not supply a useful numerical threshold in n . Nor does it assert that a uniform random initial phase converges to a nonsynchronized minimum with probability tending to one: existence of a local minimum is the target.

5 Supplementary exact finite certificates

5.1 Rational approximate phases

Two supplied rational approximate phase vectors, on connected simple cubic graphs with 20 and 500 vertices, satisfy the hypotheses of Lemma 2. The standalone verifier uses only integer and rational arithmetic for acceptance. It proves existence of nearby exact minima, not that the supplied rational vectors themselves are stationary. These finite checks are supplementary, not premises of the asymptotic proof.

For a connected graph with n vertices and diameter D ,

$$\lambda_2(L_G) \geq \frac{2}{(n-1)D} > \frac{1}{nD}. \quad (18)$$

To see this, for $z \in \mathbf{1}^\perp$ use a shortest path for each vertex pair:

$$n \|z\|_2^2 = \sum_{u < v} (z_u - z_v)^2 \leq \frac{n(n-1)}{2} D \sum_{\{u,v\} \in E} (z_u - z_v)^2.$$

The verifier computes D by exact breadth-first searches and uses $\gamma = 1/(nD)$.

For a rational edge difference x , it evaluates

$$S_N(x) = \sum_{j=0}^N \frac{(-1)^j x^{2j+1}}{(2j+1)!}, \quad C_N(x) = \sum_{j=0}^N \frac{(-1)^j x^{2j}}{(2j)!}.$$

Including the next zero Taylor coefficient, Lagrange's remainder theorem gives

$$|\sin x - S_N(x)| \leq \frac{|x|^{2N+3}}{(2N+3)!}, \quad |\cos x - C_N(x)| \leq \frac{|x|^{2N+2}}{(2N+2)!}.$$

These rational enclosures hold for every real x ; they do not require an alternating-series monotonicity assumption. The supplied runs use $N = 40$. Summing sine intervals at each vertex gives an upper bound g_2 on the squared gradient norm. The decisive test is the exact rational inequality

$$g_2 < \frac{c^4 \gamma^2}{32}. \quad (19)$$

A separate test finds an edge with original absolute real phase difference between 1 and 5. Its change is at most $c/2 \leq 1/2$, so it stays strictly between 0 and 2π in absolute value, since $2\pi > 6$.

n	D	c	γ	verified conservative bound
20	5	1/4	1/100	$g_2 < 10^{-22}$
500	12	1/8	1/6000	$g_2 < 10^{-21}$

The right sides of (19) are, respectively, $1/81\,920\,000$ and $1/4\,718\,592\,000\,000$. All acceptance comparisons use `fractions.Fraction`; decimal log entries are descriptive only. Floating-point optimization and Newton refinement were used only to propose rational inputs to the verifier.

5.2 A symbolic finite realization

There is also a finite realization of the tree profile with an exactly zero gradient, requiring no perturbation argument. Start with the four cycle roots of signs $+, +, -, -$, and build their outward trees through depth $R - 1$. At depth R , pair each prospective leaf of the first positive tree with a leaf of the first negative tree, and do the same for the other two trees. Identify each opposite-sign pair into a single vertex of phase zero. Every such vertex has two parents of opposite phases and therefore has zero net sine force. Match the resulting zero-phase vertices between the two pairing groups, giving each one a third edge.

The resulting connected simple cubic graph has $3 \cdot 2^R$ vertices. The phase expressions are integer linear combinations of $\delta_0, \dots, \delta_{R-1}$. Every edge difference is one of $0, \pm\delta_j$, or $\pm 2a_0 = \pm(2\pi - \delta_0)$. Consequently, each sine contribution is an exact rational multiple of t_R , and all gradient coordinates vanish. Every edge cosine is positive, so (4) proves strict local minimality modulo rotation. The cross-root gap is $\delta_0 > \pi/6$, so the state is nonsynchronized.

The supplied `analytic_gadget.py` constructs this graph and checks its degree, simplicity, connectivity, symbolic edge types, and all sine-flow balances using integers and rational numbers. At $R = 4$ it has 48 vertices and 72 edges; the accompanying tests also check $R = 5$. This example illustrates the same nonlinear profile but, like the other finite examples, is not a substitute for the random-graph argument.

6 Points for independent review

The proof has four distinct steps: a scalar equation supplies the positive cosine margin; the sign pattern and tree profile cancel all interior forces; compact-ball minimization corrects the boundary residual despite arbitrary exterior connections; and the fixed-cycle Poisson limit yields (17). In particular, numerical stationarity tolerances, finite examples, and unproved independence assumptions are not used to infer an asymptotic result. Independent review should check the analytic chain and the exact hypotheses of the two random-graph inputs, rather than only rerun the finite verifier.

References

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